



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2023-24(Even Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

END SEMESTER

Semester: VI

Programme: B. Tech Information Technology

Course: Artificial Intelligence and Machine Learning

Time: 3 hours

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33602	Course Outcome
CO1	Identify knowledge representation, problem solving and learning methods of AI.
CO2	Discuss knowledge and reasoning and its representation in AI.
CO3	Describe artificial intelligence, machine learning and deep learning
CO4	Analyze supervised learning algorithms and decision trees.
CO5	Evaluate machine learning models using cross-validation and sampling techniques like bootstrapping.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define Intelligent Agent .	CO1	2	2
b.	What is Forward Chaining ?	CO2	2	2
c.	Define Supervised Learning .	CO3	3	2
d.	What is K-NN algorithm ?	CO4	2	2
e.	Define PCA .	CO5	3	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Types of Agents with diagram.	CO1	2	5

b.	Describe Problem-Solving Agent and its working.	CO1	4	5
c.	Explain Production System and its characteristics.	CO1	3	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Bayes' Theorem with numerical example.	CO2	4	5
b.	Describe Rule-Based System architecture.	CO2	3	5
c.	Compare Forward vs Backward Reasoning with examples.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Types of Machine Learning with real-life examples.	CO3	2	5
b.	Differentiate Classification vs Regression problems .	CO3	3	5
c.	Explain Reinforcement Learning with agent-environment interaction.	CO3	3	5
Que. 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain CNN architecture and layers .	CO4	2	5
b.	Explain Backpropagation algorithm in neural networks.	CO4	4	5
c.	Describe Support Vector Machine (SVM) with margin concept.	CO4	2	5
Que. 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Principal Component Analysis (PCA) .	CO5	2	5
b.	Describe K-Means clustering algorithm with steps.	CO5	3	5
c.	Explain K-Fold Cross Validation and Bootstrapping .	CO5	2	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create